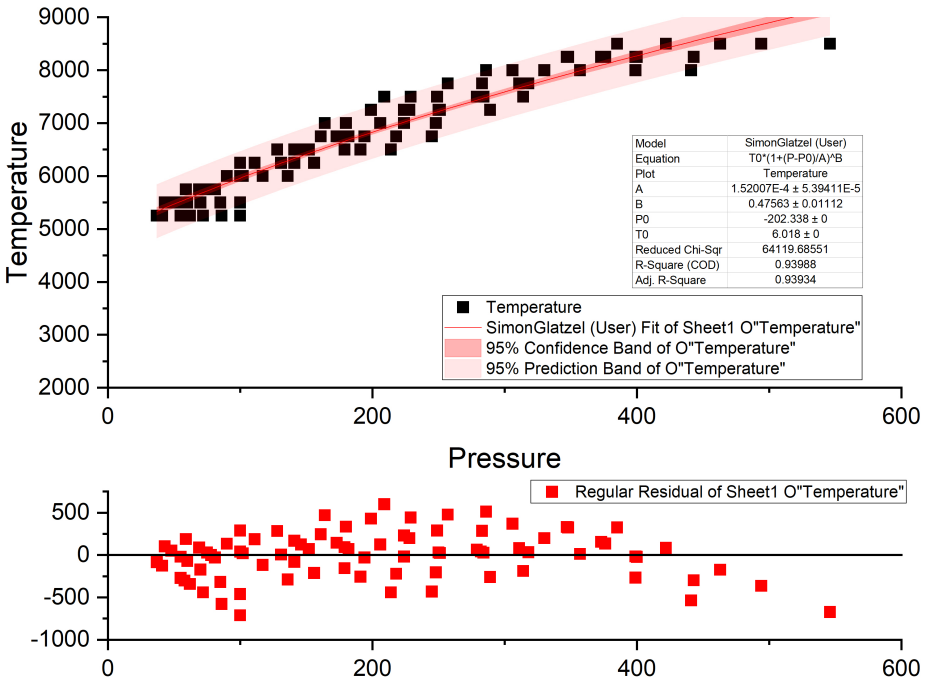
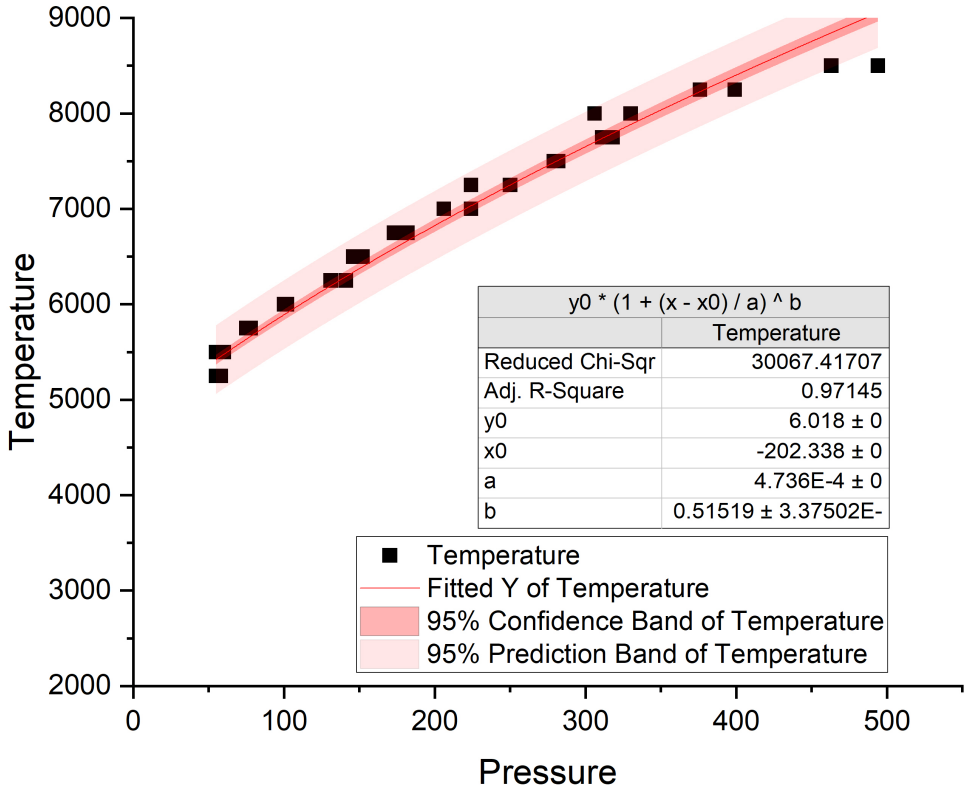
Subject: Re: Request for paper data points
Date: Tuesday, August 5, 2025 at 8:35:08 PM Mountain Standard Time
From: Ming Geng(耿明)
To: Sarah Stewart
Attachments: SimonGlatzel (User) Fit.jpg, Graph6.jpg, image.png

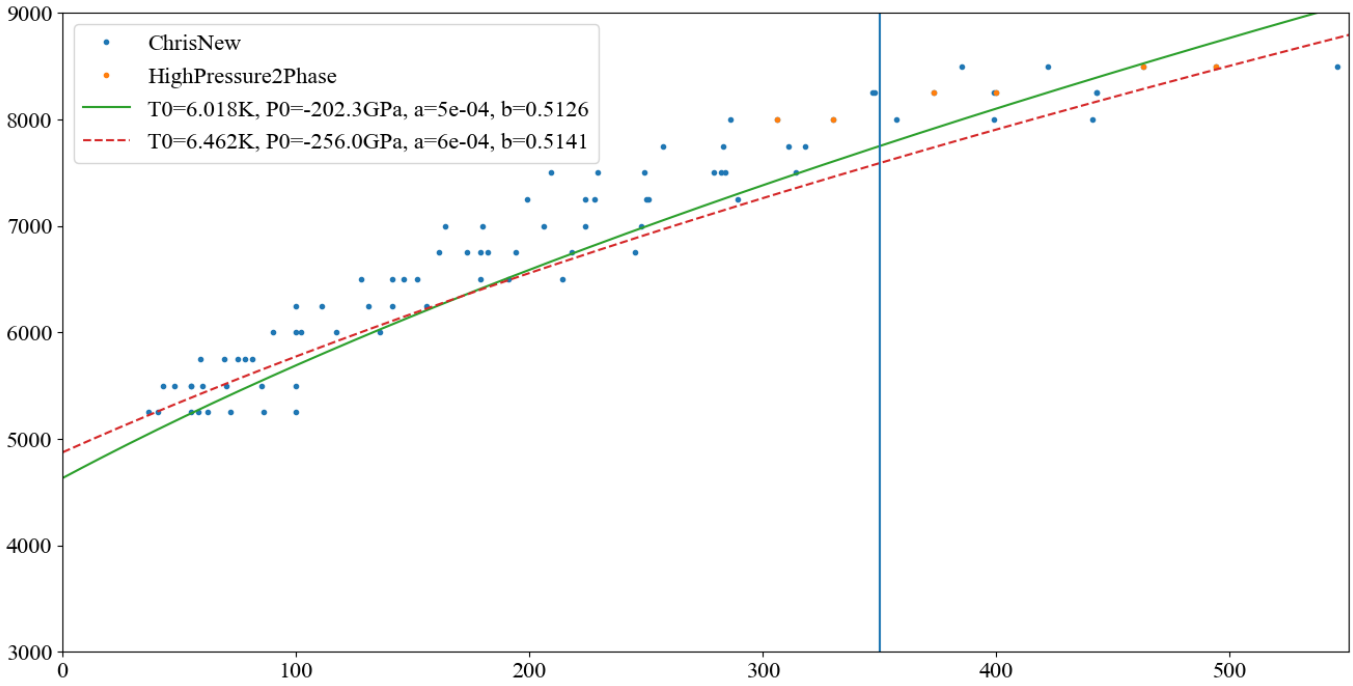
Hi Sarah,
Sorry for the late reply.

Actually, the envelope in Fig. 1 is a fitting product generated using the OriginLab software's built-in function. The darker one is the 95% confidence Band, and the lighter one is the 95% prediction band. I just found that the plot legends have the wrong description. Sorry for the confusion. The license of the software was provided by UiO, and I am not working there anymore. So I do not have the software right now. I can not reproduce the band data for you. But I found some old plots when I was trying the fitting. Some error information is shown together with the plots.





Because the original simulations are not evenly equilibria into liquid and solid, a selected pair of data was used for the final fitting. From my personal view, the Simon-Glatzel parameters we give in the paper already deviate from their supposed physical meaning (reference temperature and pressure). We just provided numerical solutions from the fitting algorithms for people to easily reproduce a curve within a reasonable error. Since the melting curve fitting from the Z-method and 2-phase simulations have some overlap region, the fitted curves should be really close to each other. I plotted the two curves with our 2-phase simulation data points as follows. I personally prefer the green line, which is the 2-phase curve reported in the paper.



Yes, the density data points were not picked exactly on the melting curve because we can not foresee the melting curve before the simulation. And all the density data were picked from the simulations that ended at the liquid state. In this way, some of the results will have a higher temperature above the melting curve.

Hope this will help you. Let me know if you have anything more you want to know.

Best regards,
Ming

Dr. Ming Geng
Computational Geochemistry

Email: gengmingchina@gmail.com

On Thu, Jul 31, 2025 at 11:10 PM Sarah Stewart <sstewa56@asu.edu> wrote:

> I check the other equation in the caption.
> Is the 2phase fit really the one labeled Z method?
> pg24arr = np.arange(500)+50 # GPa
> tg24arr = 6.462*np.power((1+(pg24arr+256.011)/6.474e-04),0.5141)
>
> Thanks,
> Sarah
>
> Using voice dictation. Apologies for typos.
>
> **From:** Sarah Stewart <sstewa56@asu.edu>
> **Date:** Thursday, July 31, 2025 at 2:04 PM
> **To:** Ming Geng(耿明) <gengmingchina@gmail.com>
> **Subject:** Re: Request for paper data points
>
> Hi Ming,
> I have a couple of questions:
> In your Figure 1, the purple line that is the 2phase fit does not quite match the equation in the caption.
> In my plot, the intercept on the right is >9000 K instead of below. See attached.
> pg24arr = np.arange(500)+50 # GPa
> tg24arr = 6.018*np.power((1+(pg24arr+202.338)/4.736e-04),0.5126)
> Is this the correct equation?
>
> Also, please confirm that the liquid densities are calculated as somewhat higher temperatures than your 2phase fit P-T melt curve.
> Your error envelopes will still be very useful.
> Thank you,
> Sarah
>
> Using voice dictation. Apologies for typos.
>
> **From:** Ming Geng(耿明) <gengmingchina@gmail.com>
> **Date:** Sunday, July 27, 2025 at 6:40 PM
> **To:** Sarah Stewart <sstewa56@asu.edu>
> **Subject:** Re: Request for paper data points
>
> Hi Sarah,
> The attached files are the original Graph and the data in the plots. Hope this helps.
>
> Best regards,
> Ming
> ---
> **Dr. Ming Geng**
> Computational Geochemistry
>
> Email: gengmingchina@gmail.com
>
> On Fri, Jul 25, 2025 at 8:38 PM Sarah Stewart <sstewa56@asu.edu> wrote:
>
> > Thank you!
> > I have done a detailed comparison between your calculations and lab experiments on SiO2 to resolve the density offset between GGA and the lab. I'm using scanned points now, but it would be much better to have your original calculation values.
> > Cheers,

Sarah

From: Ming Geng(耿明) <gengmingchina@gmail.com>
Date: Thursday, July 24, 2025 at 8:06 PM
To: Chris Erik Mohn <c.e.mohn@smn.uio.no>
Cc: Sarah Stewart <sstewa56@asu.edu>
Subject: Re: Request for paper data points

Hi Chris and Sarah,

I am now at a High pressure study conference. I just briefly checked my laptop with me and didn't find the origin data on my laptop. They must in one of my backup hard disks. I will try finding it out as soon as possible and send to Sarah.

Best Regards,
Ming

Dr. Ming Geng
Computational Geochemistry

Email: gengmingchina@gmail.com

Chris Erik Mohn <c.e.mohn@smn.uio.no> 于 2025年7月25日周五 06:02写道:

> Hi Sarah,
>
> Sure, we can provide the data-points.
>
> @Ming, I guess you have all data-points when you made the Fig ?
>
> Cheers
>
> Chris
>
> ---
>
> **From:** Sarah Stewart <sstewa56@asu.edu>
> **Sent:** Thursday, July 24, 2025 23:55
> **To:** Chris Erik Mohn <c.e.mohn@smn.uio.no>
> **Subject:** Request for paper data points
>
> Dear Professor Mohn,
>
> I am using your results from:
>
> Geng, M., & Mohn, C. E. (2024). Ab initio constraints on the melting of silica at high pressures up to 500 GPa. *Physical Review B*, *109*(2), 024106. https://doi.org/10.1103/PhysRevB.109.024106
>
> I would like to re-use the data points in Figure 2A for
> Pressure - density of the liquid
> Pressure - coordination number of the liquid
> These data points are not tabulated. I am hoping you could provide the values rather than me digitizing the figure.
>
> Thank you for your trouble,
> Sarah
>
> Professor and Associate Director for Graduate Initiatives
> School of Earth and Space Exploration, Arizona State University
> sts@sarahtstewart.net • sstewa56@asu.edu
> sarahtstewart.net impacts.wiki